

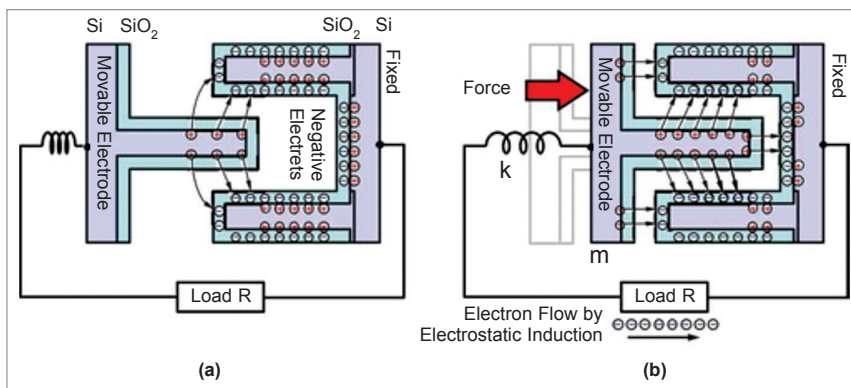


Fig. 2: Electrostatic current being generated by comb-shaped electrodes

and their relative motion dictates the amount of electrical power generated.

One of this device's uniqueness is that even the single-coil and a magnet pair can be operated in two different configurations, depending on the direction of relative displacement.

Piezoelectric devices. Piezoelectric materials are very popular in converting mechanical vibrations into electrical energy. The piezoelectric effect occurs when mechanical strain is applied to piezoelectric materials. These materials get electrically polarised under such conditions in proportion to applied strain. The piezoelectric based energy harvesters have higher energy density for all practical low-power applications.

RF energy harvesters

Radio frequency (RF) is an abundant source, and the concept of harnessing it is quite simple. Radio waves traveling in air reach the antenna and cause potential difference across its length. The RF-DC integrated circuits capture energy from the movement of those charge carriers. The energy can be stored temporarily in a capacitor, which can then be delivered across the load.

An RF-DC converter circuit generally consists of an antenna that captures RF signals, a matching circuit that matches the antenna impedance with the receiver for maximum power transfer, a voltage multiplier circuit that boosts the input voltage to a compatible value, and power conditioning circuits like low-pass filters that ensure smooth output voltage.

Recently launched power management ICs by e-peas feature an innovative buck-boost architecture that helps devices to deal with low voltages. It

targets RF-harvesting applications. It can harvest energies as low as 275mV/3μW from the environment.

Researchers at the National University of Singapore and Tohoku University have discovered a way to use magnetic tunneling junctions (MTJs) for harvesting 2.4GHz RF energy. The MTJs consist of two magnetic materials separated by around 1nm. These junctions act as a high frequency oscillator when a voltage bias is applied. With this, the researchers were able to harvest 850nW of power without any bias, which is enough to power a red LED via an LTC3108 DC-DC converter.

Triboelectric nanogenerators

These devices are the best option if your design being worked on consists of a material where friction is involved, and 100 to 200μW of power is sufficient. The triboelectric generators harvest all sorts of wasted mechanical energy that is readily available to us due to actions such as walking, rotating tires, vibration, wind, flowing water, human movement, etc. These devices can also be used as self-powered sensors to detect changes arising from

mechanical agitation.

The working of these nanogenerators is relatively simple. When a mechanical stress is applied onto this device to bend it, the inner surfaces of the two sheets get into close contact and the charge transfer takes place. Due to electrical induction, one side of the surface attracts all positive charges while the other side attracts negative charges. This is called the triboelectric effect. When the stress is released, the surfaces separate and the opposite triboelectric charges generate an electric field in between, inducing a potential difference across the top and bottom electrodes.

Thermoelectric harvesters

Thermoelectricity refers to the direct conversion of heat into electricity and vice versa. Applications that are being worked on to utilise the thermoelectric harvesters involve environments that contain temperature gradients and heat flow. These are solid-state devices that are heavily doped for high current generation. These harvesters can be employed in wearable devices for harvesting energy from body heat as well.

Recent studies suggest that transverse thermoelectricity can simplify energy harvesting from heat as the electric field and heat gradient are perpendicular. Researchers recently developed a crystal of rhenium and silicon that can produce electricity with notable efficiency at right angles to the heat flowing through it.

What to used where?

Summarising all the discussed harvesting techniques into three classes based on their feasibility, we can say the first class of systems work only when radiation coming from an on-board antenna is used. These are feasible for RF and microwave systems, but also harness ambient RF signals. An example of such a system would be wireless information and energy transfer (WIET) like the passive RFID backscatter or 6G communications at terahertz far infrared.

We can categorise the second class of systems as self-powered systems that rely on static electricity

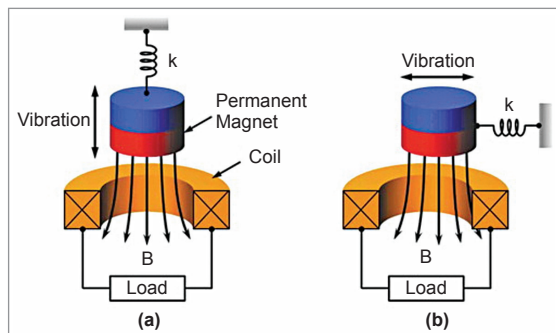


Fig. 3: Two different arrangements for relative displacement between coil and magnet: (a) magnet moving parallel to the axis of the coil, (b) magnet moving perpendicular to the axis of the coil